

## 27 Supersymmetric Gauge Theories

### 27.7 Soft Supersymmetry Breaking<sup>4</sup>

We will see in the next chapter that, even if supersymmetry is an exact symmetry of the action, the spontaneous breakdown of supersymmetry at very high energy can produce superrenormalizable terms that violate supersymmetry conservation in the effective action that describes physics at lower energies. These superrenormalizable terms may explain the lack of supersymmetry observed in phenomena at accessible energies. In this section we will consider the radiative corrections that can be produced by such supersymmetry-breaking superrenormalizable terms, in part to see whether this provides a criterion for including or rejecting such terms in supersymmetric versions of the standard model.

The sign of supersymmetry breaking is the appearance of expectation values of  $D$ -terms of general superfields or  $\mathcal{F}$ -terms of chiral superfields. Any operator  $\epsilon\mathcal{O}$  in the Lagrangian density that breaks supersymmetry can be written in a supersymmetric form, as a  $D$ -term

$$\epsilon\mathcal{O} = [ZS]_D, \quad (27.7.1)$$

where  $S$  is a non-chiral superfield that has  $\mathcal{O}$  as its  $C$ -term, and  $Z$  is a non-chiral external superfield whose only non-vanishing component is  $[Z]_D = \epsilon$ . Some but not all operators  $\epsilon\mathcal{O}$  that break supersymmetry can also be written as  $\mathcal{F}$ -terms,

$$\epsilon\mathcal{O} = [\Omega O]_{\mathcal{F}}, \quad (27.7.2)$$

or their adjoints, where  $O$  is the left-chiral superfield whose  $\mathcal{F}$ -term is  $\mathcal{O}$  and  $\Omega$  is an external left-chiral superfield whose only non-vanishing component is  $[\Omega]_{\mathcal{F}} = \epsilon$ . We can count the order in  $\epsilon$  in which a given correction to the effective Lagrangian will occur by counting the powers of  $Z$  or  $\Omega$  needed to construct this correction in a supersymmetric way. We will find interesting limitations on the radiative corrections that can be produced by those interactions that can be written in the form (27.7.2) as well as (27.7.1).

According to the results of the previous section, there are no radiative corrections to  $\mathcal{F}$ -terms, so all supersymmetry-breaking radiative corrections to the Wilsonian Lagrangian density must take the form of  $D$ -terms. This theorem does not prevent any given operator from appearing in the Wilsonian Lagrangian density, because even if an operator  $\epsilon\Delta\mathcal{L}$  cannot be expressed in the form  $[Z\Lambda]_D$ , where  $\Lambda$  is the general superfield whose  $C$ -term is  $\Delta\mathcal{L}$ , yet  $\epsilon^2\Delta\mathcal{L}$  can be expressed in the form

$$\epsilon^2\Delta\mathcal{L} = 2[\Omega^*\Omega\Lambda]_D. \quad (27.7.3)$$

But not all operators can be produced by radiative corrections of *first order* in  $\Omega$  or  $\Omega^*$ . In particular, a function only of the  $\phi$ -term of a left-chiral superfield  $\Phi$ , but not of  $\phi^*$ , cannot be written as the  $D$ -term of a superfield linear in  $\Omega$ . (Note that  $[\Omega h(\Phi)]_D$  is a derivative, while  $[\Omega^* h(\Phi)]_D = 2[\Phi]_{\mathcal{F}} \partial h(\phi)/\partial\phi$  is not a function of  $\phi$  alone.) We conclude

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<sup>4</sup>This section lies somewhat out of the book's main line of development and may be omitted in a first reading.

then that *the supersymmetry-breaking terms in the Wilsonian Lagrangian that depend on  $\phi$  alone cannot be produced by radiative corrections that are of first order in supersymmetry-breaking interactions of the form (27.7.2).*

This result is significant because the most divergent radiative corrections are those that are of lowest order in superrenormalizable couplings. To be more specific, the coefficient of an interaction of dimensionality  $\mathcal{D}$  has dimensionality (in powers of energy)  $4 - \mathcal{D}$ , so dimensional analysis indicates that the contribution of a set of interactions of dimensionality  $d_1, d_2$ , etc. to the coefficient of an interaction of dimensionality  $d$  can contain the ultraviolet cut-off to at most a power

$$p = 4 - d - (4 - d_1) - (4 - d_2) - \dots \quad (27.7.4)$$

and is therefore finite if  $p < 0$ . (This argument ignores possible ultraviolet divergences in subintegrations; for a thorough treatment of this topic, see Reference 8.) Superrenormalizable interactions are ‘soft,’ in the sense that they reduce the degree of divergence of the graphs in which they appear. In particular, in a renormalizable theory where all interactions have  $d_i \leq 4$ , and the strictly renormalizable interactions with  $d_i = 4$  are supersymmetric, the contribution of one or more superrenormalizable interactions to the coefficient of an interaction with  $d = 4$  will always have  $p < 0$ , so even if they are not supersymmetric the superrenormalizable interactions will not produce supersymmetry-violating ultraviolet-divergent corrections to the coefficients of the supersymmetric  $d = 4$  interactions.

On the other hand, in such a theory there may be divergent radiative corrections to the superrenormalizable interactions themselves [9]. The most worrisome are quadratic (or higher) divergences, which if cut off at some high energy scale  $M_X$  may require a fine-tuning of the bare coupling constants to preserve supersymmetry as a good approximate symmetry at energies below  $M_X$ . According to Eq. (27.7.4), in renormalizable theories where all interactions with  $d_i = 4$  are supersymmetric, radiative corrections can produce quadratic or more highly divergent ( $p \geq 2$ ) supersymmetry-violating operators of dimensionality  $d$  only if they involve insertion of a superrenormalizable supersymmetry-violating interaction of dimensionality  $d_1 \geq 2 + d$ . This allows either  $d = 0$  and  $d_1 \geq 2$ , which arises only when we calculate the cosmological constant, or  $d = 1$  and  $d_1 = 3$ , which arises only when we calculate the ‘tadpole’ graphs in which a scalar field line disappears into the vacuum. The cosmological constant raises fine-tuning problems for all known theories [10] and will not be considered further here. The tadpole graphs represent operators linear in  $\phi$  or  $\phi^*$  and, as we have seen, can not be produced to first order in supersymmetry-violating interactions that can be put in the form (27.7.2). Such superrenormalizable interactions are therefore ‘soft’ in the sense that they do not induce quadratic or higher divergences. Along with the superrenormalizable interactions with  $d \leq 2$ , including arbitrary quadratic polynomials in  $\phi$  and  $\phi^*$ , the supersymmetry-breaking interactions that are soft in this sense include terms of third order in the  $\phi$ , which can be expressed as  $\phi^3 = [\Omega\Phi^3]_{\mathcal{F}}$ , and likewise terms of third order in the  $\phi^*$ , and also  $d = 3$  gaugino mass terms, which can be written as  $[\Omega\epsilon_{\alpha\beta}W_\alpha W_\beta]_{\mathcal{F}}$ , but not terms like  $\phi^2\phi^*$  or  $\phi\phi^{2*}$ , which in general *can* produce quadratically divergent tadpole graphs [9].

Nevertheless, tadpoles can only arise for scalar fields that are neutral with respect to all exact symmetries. In theories without such neutral scalars, like the supersymmetric standard model discussed in the next chapter, *all* superrenormalizable interactions may be considered to be soft.